

Build a Professional AI-Powered Camera App for iPhone (SwiftUI + AVFoundation + CoreImage + Vision + Metal)

Project Overview

Create a premium professional photography application for iPhone that combines DSLR-level manual controls, computational photography, RAW editing, AI-powered enhancements, and a modern one-handed user experience.

The app should feel like a combination of Halide, Lightroom Mobile, Luminar Neo, and Apple's native Camera app while maintaining a clean, intuitive Apple Human Interface Guidelines (HIG) design.

Technical Requirements

Platform

- iOS 18+
- Swift 6
- SwiftUI
- AVFoundation
- CoreImage
- Metal Performance Shaders
- Vision Framework
- CoreML
- Photos Framework
- PencilKit (for masking/editing)
- StoreKit
- Combine
- Async/Await

Architecture

- MVVM
- Modular Architecture
- Dependency Injection
- Repository Pattern
- SwiftData persistence
- Background processing pipeline
- GPU accelerated image processing

Camera Features

Capture Modes

Photo

- JPEG
- HEIF
- RAW (DNG)
- Apple ProRAW

Portrait

- Depth capture
- Bokeh adjustment
- Edge refinement

Night Mode

- Long exposure stacking
- Noise reduction
- AI detail preservation

Macro Mode

- Auto lens switching
- Focus assistance

Panorama

- Live horizon guidance

Burst Mode

- Intelligent best-shot selection

Professional Camera Controls

Manual Exposure

Provide DSLR-style manual controls:

- ISO
- Shutter Speed

- Exposure Compensation
- EV Lock
- Exposure Bracketing

UI

- Bottom thumb wheel
 - Horizontal adjustment dial
 - Haptic feedback
 - Real-time preview
-

Manual Focus

- Focus slider
- Rack focus support
- Focus lock

Focus Peaking

- Real-time edge detection
 - Highlight focused regions
 - Adjustable peaking intensity
 - Color options:
 - Red
 - Blue
 - Yellow
 - Green
-

White Balance

Custom controls:

- Kelvin slider
- Tint slider

Presets:

- Auto
 - Daylight
 - Cloudy
 - Shade
 - Tungsten
 - Fluorescent
 - Flash
-

Histogram

Real-time histogram overlays:

- RGB Histogram
- Luminance Histogram

Options:

- Compact view
 - Expanded view
 - Hide/show
-

Grid Overlays

Support:

- Rule of Thirds
- Golden Ratio
- Golden Spiral
- Triangle Composition
- Square Grid
- Custom Grid

Opacity controls.

RAW & ProRAW Workflow

RAW Capture

Capture:

- DNG
- ProRAW

Preserve:

- Wide color gamut
- Extended dynamic range
- 12-bit and 14-bit data

Editing should remain fully non-destructive.

Completely Redesigned One-Handed UI

Design Goal

Every critical camera control must be accessible using only the thumb.

No deep settings menus.

No more than 2 taps to access any major feature.

Layout

Top Bar

Contains:

- Battery indicator
 - Storage indicator
 - Flash
 - Live Photo
 - RAW/ProRAW toggle
 - Settings
-

Center Viewfinder

Full-screen camera preview.

Optional overlays:

- Histogram
 - Grid
 - Focus Peaking
 - Level Indicator
 - Horizon Guide
-

Thumb Control Zone

Bottom 35% of screen dedicated to one-handed operation.

Contains:

- Exposure wheel
- Focus wheel
- White balance wheel
- Zoom control
- Capture button

Swipe left/right:

- Switch controls

Long press:

- Quick presets
-

Quick Access Control Dock

Floating dock above shutter button:

Buttons:

- ISO
- Shutter
- Focus
- WB
- Histogram
- Grid
- AI Tools

Customizable by user.

Gesture System

Single tap:

- Focus

Double tap:

- Exposure lock

Long press:

- Focus + exposure lock

Swipe up:

- Show advanced controls

Swipe down:

- Hide controls

Pinch:

- Zoom

Two-finger swipe:

- Lens switch
-

Built-In Professional Photo Editor

Editing Philosophy

Non-destructive editing pipeline.

Every edit stored as metadata.

Unlimited undo/redo.

Develop Core Module

Manual adjustments:

- Exposure
- Contrast
- Highlights
- Shadows
- Whites
- Blacks
- Brightness
- Saturation
- Vibrance
- Temperature
- Tint
- Clarity
- Texture

- Sharpness
- Noise Reduction
- Vignette

Real-time GPU preview.

Curves Module

Professional tone curves.

Support:

- RGB Curve
- Red Curve
- Green Curve
- Blue Curve

Features:

- Curve presets
 - Touch editing
 - Histogram overlay
-

Detail Module

Controls:

- Sharpening
- Radius
- Detail
- Masking
- Noise Reduction
- Luminance Reduction
- Color Noise Reduction

Multi-level sharpening pipeline.

Looks System

One-tap professional looks.

Categories:

Natural

- Balanced
- Clean
- Soft

Cinematic

- Teal & Orange
- Hollywood
- Drama

Portrait

- Skin Soft
- Editorial
- Beauty

Landscape

- Vivid Nature
- Golden Hour
- Mountain Air

Urban

- Street
- Neon
- Night City

Adjust intensity with slider.

Specialty Filters

Golden Hour

Applies:

- Warm sunlight gradients
- Golden highlights
- Soft shadow lift

Adjustable strength.

Foliage Enhancer

Detects vegetation.

Boosts:

- Green saturation
 - Vibrance
 - Leaf texture
 - Micro contrast
-

Dehaze & Fog Removal

Controls:

- Atmospheric haze reduction
 - Fog removal
 - Contrast recovery
 - Sky restoration
-

Monochrome & Film Simulation

Black & White:

- Classic Monochrome
- High Contrast
- Fine Art

Film Emulations:

- Kodak Portra
- Kodak Gold
- Fuji Provia
- Fuji Velvia
- Ilford HP5
- CineStill

Grain controls:

- Size

- Roughness
 - Intensity
-

AI Editing Engine

Sky AI

Automatically:

- Detect horizon
- Segment sky

Features:

- Sky replacement
- Sunset skies
- Storm skies
- Blue skies
- Dramatic clouds

Controls:

- Blend
 - Warmth
 - Brightness
 - Reflection matching
-

Enhance AI

Single intelligent slider.

Balances:

- Exposure
- Shadows
- Highlights
- Contrast
- Saturation

Produces natural results.

Structure AI

Enhances:

- Texture
- Fine detail
- Clarity

Avoids:

- Halo artifacts
 - Excessive sharpening
 - Digital noise
-

Relight AI

Create a virtual 3D light map.

Capabilities:

- Foreground brightening
- Background brightening
- Face illumination
- Subject relighting

Controls:

- Direction
 - Intensity
 - Radius
 - Warmth
-

Erase Tool

Content-aware removal.

Remove:

- Photobombers
- Sensor dust
- Wires
- Small distractions

Brush controls:

- Size
 - Feather
 - Opacity
-

Crop & Transform

Support:

- Straighten
- Rotate
- Perspective correction
- Vertical correction
- Horizontal correction

Aspect Ratios:

- Freeform
 - Original
 - 1:1
 - 4:5
 - 3:2
 - 16:9
-

Photo Library

Professional gallery.

Features:

- RAW badges
- ProRAW badges
- AI edit badges
- Favorites
- Albums
- Smart search

Search by:

- Camera
- Lens
- Date

- Location
 - Subject
-

Performance Requirements

Camera launch:

- < 1 second

Capture latency:

- < 100 ms

RAW processing:

- GPU accelerated

Editing:

- 60 FPS preview

Memory optimized for large RAW files.

Accessibility

- Dynamic Type
 - VoiceOver
 - Haptic feedback
 - High contrast mode
 - Left-handed mode
 - One-handed mode
-

Premium UX Details

- Liquid Glass style effects
- Smooth transitions
- Subtle haptics
- Real-time control animations
- Contextual tooltips
- Onboarding tutorial
- Customizable quick controls
- Dark Mode

Deliverables

Generate:

1. Complete SwiftUI application
2. MVVM architecture
3. AVFoundation camera engine
4. RAW + ProRAW pipeline
5. Metal accelerated editor
6. AI editing modules
7. Custom camera UI
8. Unit tests
9. UI tests
10. Documentation
11. App Store ready build configuration

The final app should be capable of competing with Halide, Lightroom Mobile, ProCamera, and Luminar Mobile while maintaining Apple's design standards and exceptional one-handed usability.